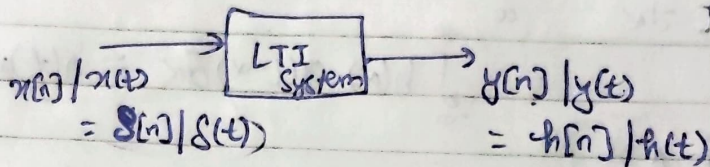


Linear Time Invariant System

For linearity and time invariance, the output signal must be a weighted superposition of time shifted impulse responses. This weighted superposition is termed as convolution sum for DT System and convolution integral for CT system.



$h(t) \rightarrow$ Impulse f^n
 $H(s) \rightarrow$ Transfer f^n
 $H(j\omega) \rightarrow$ Frequency Response

<u>Input</u>	<u>Output</u>	
$\delta[n]$	$h[n]$	
$a \cdot \delta[n]$	$a h[n]$	$\rightarrow$ Homogeneity
$\delta[n-k]$	$h[n-k]$	$\rightarrow$ TI

$x[k] \cdot \delta[n-k] \quad x[k] \cdot h[n-k] \quad \rightarrow$ Homogeneity + TI

$x[k] \cdot \delta[n-p] + x[k] \cdot \delta[n-k] \quad x[p] \cdot h[n-p] + x[k] \cdot h[n-k] \quad \rightarrow$ Linearity

$x[n] = \sum_{k=-\infty}^{\infty} x[k] \cdot \delta[n-k] \quad y[n] = \sum_{k=-\infty}^{\infty} x[k] \cdot h[n-k] \quad \rightarrow$ Convolution sum

$x(t) = \int_{-\infty}^{\infty} x(\tau) \delta(t-\tau) d\tau \quad y(t) = \int_{-\infty}^{\infty} x(\tau) h(t-\tau) d\tau \quad \rightarrow$ Convolution Integral

$h(t) / h(n) \rightarrow$ Completely characteristic / represents the CT, DT LTI System
 $\downarrow$
 Impulse Response

* Convolution sum/integral also shows commutative property

Proof $\Rightarrow x(t) * h(t) = \int_{-\infty}^{\infty} x(\tau) h(t-\tau) d\tau$

$$\begin{aligned} \text{let } t-\tau = \alpha & \quad = - \int_{\infty}^{-\infty} x(t-\alpha) \cdot h(\alpha) d\alpha \\ d\tau = d\alpha & \\ & = \int_{-\infty}^{\infty} h(\alpha) \cdot x(t-\alpha) d\alpha = h(t) * x(t) \end{aligned}$$

$x * h = h * x$

* Convolution sum/integral also shows associative property

Proof $\Rightarrow x_3(t) * [h_1(t) * h_2(t)] = x_3(t) * \left[\int_{-\infty}^{\infty} h_1(\tau) \cdot h_2(t-\tau) d\tau \right]$

$$\begin{aligned} &= \int_{-\infty}^{\infty} x(\alpha) \cdot \left[\int_{-\infty}^{\infty} h_1(\tau) \cdot h_2(t-\tau-\alpha) d\tau \right] d\alpha \\ \text{let } \tau-\alpha &= \beta \quad d\tau = d\beta \\ &= \int_{-\infty}^{\infty} x(\alpha) \left[\int_{-\infty}^{\infty} h_1(\beta-\alpha) h_2(t-\beta) d\beta \right] d\alpha \\ &= \int \int x(\alpha) h_1(\beta-\alpha) h_2(t-\beta) d\beta d\alpha \\ \text{Interchanging inside} &= \int \left[\int x(\alpha) h_1(\beta-\alpha) d\alpha \right] h_2(t-\beta) d\beta \\ &= \int [x(\beta) * h_1(\beta)] \cdot h_2(t-\beta) d\beta \\ &= [x(\beta) * h_1(\beta)] * h_2(\beta) \end{aligned}$$

$[x * h_1] * h_2 = x * [h_1 * h_2]$

* Distributive property ✓

* Time shifting ✓

* Sum Property (Only for discrete)

S_x, S_y, S_h are sum of all impulses ~~equal~~ in $x(n), y(n), h(n)$
 then $S_y = S_x \cdot S_h$ where $y(n) = x(n) * h(n)$

Width property

* If duration of $x(t)$ & $h(t)$ are w_x & w_h then
 $x * h$ duration is $w_x + w_h$. Also edges are added too.

For discrete time: $(N_x + N_h - 1)$ are samples in $x(n) * h(n)$

* Differentiation prop

$$x * h = y \rightarrow \boxed{x' * h = x * h' = y'}$$

* Time scaling

$$\boxed{x(at) * h(at) = \frac{1}{|a|} y(at)}$$

Ex 1 $x(n) = e^{-n^2}$, $h(n) = 3n^2 \forall n$ find $x(n) * h(n)$
 $\Rightarrow$

$$y(n) = x(n) * h(n) = \sum_{k=-\infty}^{\infty} e^{-k^2} \cdot 3(n-k)^2$$

$$= \sum_k e^{-k^2} \cdot 3[n^2 + k^2 - 2nk]$$

$$= \sum_k 3n^2 \cdot e^{-k^2} + \sum_k k^2 \cdot 3e^{-k^2} - \sum_k 3e^{-k^2} \cdot 2nk$$

$$= 3n^2 \sum_k e^{-k^2} + 3 \sum_k k^2 e^{-k^2} - \underbrace{6n \sum_k k \cdot e^{-k^2}}_0$$

$$= \underline{\underline{5.38n^2 + 2.654}} \quad \forall n$$

* Ex 2 $u(t) * u(t)$

$$\Rightarrow \int_{k=-\infty}^{\infty} u(k) \cdot u(t-k) \cdot dk$$

$$u(k) = \begin{cases} 0 & : k < 0 \\ 1 & : k > 0 \end{cases}$$

$$\Rightarrow \int_0^{\infty} 1 \cdot u(t-k) dk$$

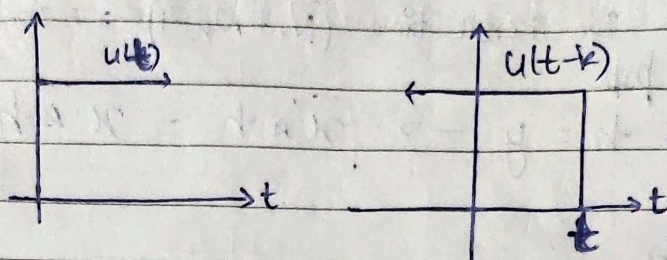
$$u(t-k) = \begin{cases} 1 & : k < t \\ 0 & : k > t \end{cases}$$

$$\Rightarrow \int_0^t 1 dk = t \quad (t > 0)$$

$$\Rightarrow t u(t) = \delta(t)$$

Ex 2) $u(t) * u(t)$ by graphical method

$$\Rightarrow \int_{-\infty}^{\infty} u(k) \cdot u(t-k) dk = y(t)$$



Case 1: $t < 0$, $y(t) = 0$

Case 2: $t > 0$, $y(t) = \int_0^t 1 \cdot dk = t$

Case 3: $t = 0$, $y(t) = 0$

* Some properties $\rightarrow$

$$\begin{cases} \bullet \delta(t) * u(t) = u(t) \\ \bullet u(t) * u(t) = tu(t) = r(t) \end{cases}$$

$[n]u[n] = r[n]$

$$\begin{cases} \bullet \delta[n] * u[n] = u[n] \\ \bullet u[n] * u[n] = (n+1)u[n] = r[n+1] \end{cases}$$

* If $x(t) * y(t) = z(t)$
 then $x(t-t_0) * y(t-t_1) = z(t-t_0-t_1)$
 $= x(t) * y(t-t_0-t_1)$

For $u(t-t_1) * u(t-t_2) = (t-t_1-t_2)u(t-t_1-t_2)$

$\bullet x[n] * \delta[n] = x[n]$

$\bullet x(t) * \delta(t) = x(t)$

- In cascade interconnection, left limit is $T_1 + \delta$,
- In parallel " " " " $\min(T_1, \delta_1)$

where T_1, δ, δ_1 are lower limits of 2 systems

Proof: $\Rightarrow z(t) = x(t) * y(t) = \int_{-\infty}^{\infty} x(l) \cdot y(t-l) dl$ - (1)

$\Rightarrow$ Let $x(t-t_0) * y(t-t_1) = \int_{-\infty}^{\infty} x(l-t_0) \cdot y(t-t_1-l) dl$

$\Rightarrow = \int_{-\infty}^{\infty} x(\alpha) \cdot y(t-t_1-t_0-\alpha) d\alpha$ - (2)

From eqn (1) & (2) $\Rightarrow$

$$\boxed{x(t-t_0) * y(t-t_1) = z(t-t_1-t_0)}$$

(Ex 2) $z(t) = x(t) * y(t)$. find $x(2t) * y(t)$ in $z(t)$

soln: $z(t) = \int_{-\infty}^{\infty} x(l) \cdot y(t-l) \cdot dl$ - (1)

$\Rightarrow x(2t) * y(t) = \int_{-\infty}^{\infty} x(2l) \cdot y(t-l) dl$
 $= \frac{1}{2} \int_{-\infty}^{\infty} x(m) \cdot y(t-\frac{m}{2}) \cdot dm$ - (2)

From eqn (1) & (2);

$$x(2t) * y(t) = \frac{1}{2} z(t)$$

* [Note]: As $x(t) * u(t) = \int_{-\infty}^t x(\tau) d\tau$ $\therefore$ an integrator is an LTI system with impulse response $u(t)$

$x(t) \rightarrow \boxed{u(t)} \rightarrow z(t)$

→ Causality for LTI system

$$y(n) = \sum_{k=-\infty}^{\infty} h(k)x(n-k) = \sum_{k=-\infty}^{-1} h(k)x(n-k) + \sum_{k=0}^{\infty} h(k)x(n-k)$$

for discrete time causal system;

$$h(k) = 0 \text{ for } k < 0$$

∴ for $k \neq 0 \rightarrow x(n+1), x(n+2), \dots$

requires future value of input

for causality; $y(n) = \sum_{k=0}^{\infty} h(k)x(n-k)$

$$= \sum_{k=-\infty}^n h(k)x(n-k)$$

(Same for CT)

⇒ Stability

$$y(n) = \sum_{k=-\infty}^{\infty} h(k)x(n-k)$$

$$|y(n)| \leq \sum_{k=-\infty}^{\infty} |h(k)||x(n-k)| \rightarrow \sum_{k=-\infty}^{\infty} |h(k)| < \infty$$

$$|y(n)| \leq |h(k)| B_x = B_x \cdot \sum_{k=-\infty}^{\infty} |h(k)| \quad \downarrow \quad \text{stable}$$

∴ if impulse response is summable the $y(n)$ is bounded hence stable

(same for CT)
(integrable)

⇒ Invertibility

$$\Rightarrow x_1(t) * [h_1(t) * h_2(t)] = x(t)$$

⇒

$$s(t) = [h_1(t) * h_2(t)] \rightsquigarrow \text{for invertible system}$$

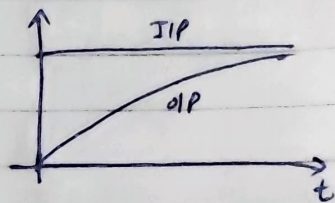
Causal $\rightarrow h(t) = 0, t < 0 \Rightarrow s(t) = 0$ for $t < 0$

Unit Step Response $\leadsto$ ^{Response} Due to a unit step signal

The step response of a DT LTI system is the running sum of its impulse response $\Rightarrow \sum_{k=-\infty}^{\infty} h(k) \cdot u(n-k) = h(n) * u(n) = s(n)$

The impulse response of a DT LTI system is first diff. of its step response $\Rightarrow \boxed{h(n) = s(n) - s(n-1)}$

$$\underbrace{h(t)}_{\text{Impulse response}} = \frac{d}{dt} \underbrace{(s(t))}_{\text{step response}}$$



Step response $s(n)$ or $s(t)$ is the output due to a unit step input signal $x(n)$ or $x(t) = u(n)$ or $u(t)$

Step response for Cont. time LTI System: $s(t) = \int_{-\infty}^t h(\tau) u(t-\tau) d\tau$
 $\downarrow$
 $s(t) = h(t) * u(t)$

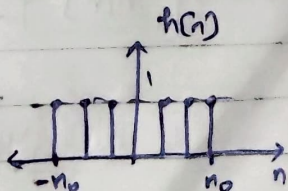
Ex: Check if LTI: $y(n) = \sum_{k=n-n_0}^{n+n_0} x[k]$

Soln: $y(n) = \sum_{k=n-n_0}^{n+n_0} x[k] \cdot 1[n-k] = x[n] * h[n]$

$$\Rightarrow h[n-k] = \begin{cases} 1 & : n-n_0 \leq k \leq n+n_0 \\ 0 & : \text{otherwise} \end{cases}$$

PTD $\rightarrow$

$$\Rightarrow h[n] = \begin{cases} 1 & : -n_0 \leq n \leq n_0 \\ 0 & : \text{otherwise} \end{cases} = \text{Rectangular function}$$



= Rectangular function
 LTI system

